Foundations for a Personal AI Learning Substrate

A Unified Internal Monograph for Pete

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Status. This document is intended for internal circulation only. It does not claim physical completeness, empirical validation, or theoretical unification of physics, cognition, and semantics. It is a working monograph for building a personal AI system whose representations, memory, meaning, identity, and semantic growth are modeled in terms of interaction rather than static symbolic storage or pretrained frozen geometry.

Abstract

This monograph presents a unified internal foundation for building a personal AI system called Pete. The framework is organized into three integrated parts. Part I, Prime Substrate, defines the ontological layer: states, interactions, difference, time, space, meaning in frame, and the energetic–material asymmetry at the heart of the system. Part II, Prime Chakra, defines the primordial semantic scaffold: a rooted set of anchor maps, seed vocabularies, and pairwise friction dynamics that allow semantic growth to begin on structured ground rather than on a flat lexical plane. Part III, Semantic Space Engine, defines the runtime learning machinery that transforms raw text into a living semantic space through situation-first preprocessing, symbol statistics, co-occurrence structure, node verification, field gravity, pre-linguistic primordials, and dynamic context embedding.

A central clarification running through the whole monograph is that Dung and Hoai are not merely symmetric abstract opposites. They are base properties expressed differently across reality. Dung is the resonant side: the property of coherence, phase alignment, constructive coupling, and the energetic tendency toward pattern reinforcement. Hoai is the decaying side: the property of erosion, friction, displacement, and breakdown that becomes most visible once matter or matter-like structure has formed. Energy tends to manifest Dung more strongly; matter tends to manifest Hoai more clearly. Stable structure arises when matter is guided into the grooves, nets, or resonant basins produced by energetic interference across time. Yet the same material embodiment also makes that structure vulnerable to collision, displacement, and breakdown. Thus, structure is never the pure essence of matter, but a temporary victory of Dung over Hoai in a given region.

This document does not argue that the world is literally composed of these categories. It argues something narrower and more useful: if a personal AI is to possess continuity, growth, memory, contextual meaning, and identity, then these capacities should be modeled structurally. The result is a complete internal design framework in which ontology, primordial anchoring, and semantic runtime are aligned into one system.

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Part I

Prime Substrate: An Interaction-Based Ontology

Chapter 1

Introduction to the Ontological Layer

Most contemporary AI systems are built from tokenization, vectorization, parametric compression, and large-scale optimization. These systems are useful, but they are not naturally organized for persistent personal cognition. A personal AI must do more than answer prompts. It must preserve identity, maintain continuity of meaning, adapt meaning across frames, remember, forget, reorganize, and grow from its own experience.

The first task, therefore, is not to build a better output layer, but to define the substrate of cognition itself. Prime Substrate is proposed as that substrate. It begins from a simple thesis:

> The primitive unit of a personal AI should not be an object, a token, or a frozen embedding. It should be a state and its interaction profile.

This ontological move matters because static objects do not explain how meaning moves, how identity persists, how memory decays, or how difference is measured. Interaction does.

A second, equally important clarification concerns Dung and Hoai. If these are treated merely as abstract positives and negatives, then the system becomes shallow and ambiguous. The intended ontology is sharper:

- **Dung** is the resonant property of reality.
- **Hoai** is the decaying property of reality.

But this is still not enough. These are *base properties*, not substances. In concrete interaction:

- energy tends to manifest Dung more strongly, because it interferes, aligns, and creates resonant patterns through time;
- matter tends to manifest Hoai more clearly, because once structure is embodied, it becomes subject to friction, displacement, collision, wear, and breakdown.

This distinction is one of the pillars of the whole monograph. It controls how we think about existence, space, memory, semantic stability, and runtime learning.

Chapter 2

Problem Statement and Design Position

2.1. Why static representation is insufficient

The dominant language model pipeline is effective for high-quality continuation, but weak as an account of personal cognition. It does not naturally answer:

- What persists when no prompt is present?
- What changes when a concept is seen from a different frame?
- What makes one memory strengthen while another weakens?
- What is the difference between resonance and mere recurrence?
- How does identity remain coherent across many local interactions?

The problem is not that token-based models are useless. The problem is that they do not provide a satisfying substrate for continuity.

2.2. Need for an interaction-first substrate

The central hypothesis of this part is:

Axiom 2.1 (Interaction primacy)**.** A system suitable for personal AI design should treat interaction type as ontologically prior to fixed object identity.

This implies:

- generation is not merely sampling but emergence,
- evaluation is not merely scoring but probing,
- memory is not mere storage but persistence under resonance,
- forgetting is not deletion but weakening under decay,
- meaning is not intrinsic to patterns but projected through frames,
- identity is not a label but a stable relational coherence.

2.3. What this framework is and is not

This framework is:

- a computational ontology,
- an internal design grammar,
- a substrate for personal AI.

It is not:

- a claim of physical truth,
- a new physics theory,
- a replacement for all current ML methods,
- a fully validated scientific model.

Chapter 3

State Formalism

3.1. State definition

Let $\mathcal{Q}$ denote the set of internal states. A state $q \in \mathcal{Q}$ is defined by:

$$q = (\mathrm{spin}(q), \mathrm{phase}(q), \mathrm{vec}(q), \mathrm{collapsed}(q)). \tag{3.1}$$

Where:

- $\mathrm{spin}(q)$ denotes interaction polarity or compatibility class,
- $\mathrm{phase}(q)$ denotes alignment angle,
- $\mathrm{vec}(q)$ denotes projection in internal space,
- $\mathrm{collapsed}(q)$ denotes whether the state is unresolved or stabilized.

3.2. Interpretation

These are computational abstractions, not literal physical claims. Their purpose is to give the system a language for:

- uncertainty,
- alignment,
- stabilization,
- exclusion,
- dynamic compatibility.

Definition 3.1 (Superposed state)**.** A state q is superposed if $\mathrm{collapsed}(q) = 0$.

Definition 3.2 (Collapsed state)**.** A state q is collapsed if $\mathrm{collapsed}(q) = 1$.

Chapter 4

The Prime Family

4.1. Overview

A prime is not a substance. It is an interaction mode. The prime family provides a minimal grammar for emergence, probing, resonance, decay, difference, time, space, and semantic framing.

4.2. Prime 0: Vo

Definition 4.1 (Vo)**.** Vo is latent potential under continued non-interaction.

Vo is not nothingness. It is dormant possibility, inactive memory, null activation, and background potential.

4.3. Prime 1: Sinh

Definition 4.2 (Sinh)**.** If A and B are unresolved states, then their interaction instantiates Sinh:

$$\mathrm{Sinh}(A, B) \sim \text{emergent novelty}.$$

Sinh is the seed of new structure, new nodes, and new unresolved possibilities.

4.4. Prime 2: Dan

Definition 4.3 (Dan)**.** If S is unresolved and C is stabilized, then

$$\mathrm{Dan}(S, C) = \text{probe of possible alignment of } S \text{ relative to } C.$$

Dan is the prime of probing, testing, attention, and pre-collapse evaluation.

4.5. Prime 3: Dung

Definition 4.4 (Dung)**.** Dung is the prime of resonance. It is the property of coherence, phase alignment, constructive coupling, and the reinforcement of pattern through coordinated interaction.

The subtle but crucial point is this: Dung is a base property. It is not identical to stable matter. It is the principle by which coherence becomes possible.

In concrete operation, energy tends to manifest Dung more strongly. Why? Because energy interferes, aligns, oscillates, and creates resonant patterns through time. Through repeated interaction, those patterns generate grooves, nets, basins, or stable channels. Matter can then be captured by those channels.

So the right claim is not:

> Dung is stable matter.

The right claim is:

> Dung is the resonant principle that makes stable matter possible.

4.6. Prime 4: Hoai

Definition 4.5 (Hoai)**.** Hoai is the prime of decay. It is the property of erosion, displacement, friction, breakdown, and destabilization that appears once structure has taken form.

Like Dung, Hoai is a base property. But its most vivid expression appears on the material side. Once matter or matter-like structure has formed, it becomes vulnerable to:

- collision,
- displacement,
- friction,
- cumulative wear,
- threshold crossing and breakage.

So Hoai is not just "disappearance." It is the principle by which stable form is gradually stressed, shifted, and potentially broken.

4.7. Prime 5: Gap

Definition 4.6 (Gap)**.** Let $A, B \in \mathcal{Q}$ and let C be an observer or frame. Then:

$$\mathrm{Gap}(A, B; C) = C(A) - C(B)_C.$$

Gap is measured difference, not just raw contrast. It requires a third term.

4.8. Prime 6: Time

Definition 4.7 (Time)**.** Time is ordered interaction sequence:

$$\mathrm{Time} = [\mathcal{P}_1 \to \mathcal{P}_2 \to \mathcal{P}_3 \to \cdots].$$

Time is not merely clock. It is ordered transformation.

4.9. Prime 7: Space

Definition 4.8 (Space)**.** Space is structured separation generated when exclusion prevents states from fully occupying the same effective condition.

Space is not frozen time, and not stacked Dung. Space belongs closer to matter and constraint. Same-like states that cannot fully merge must remain apart. Maintained separation becomes distance; patterned distance becomes space.

4.10. Prime 8: NMF

Definition 4.9 (NMF)**.** Let x be a stabilized pattern and F a frame. Then:

$$\mathrm{NMF}(x, F) = \{N(x), M(x \mid F), F\},$$

where N is the stabilized name or identity pattern and M is the meaning projected within frame F.

No meaning exists without frame.

Chapter 5

Difference, Meaning, Memory, and Identity

5.1. Difference requires three terms

Proposition 5.1. Two states alone do not yield measured difference. They yield only interaction. Difference becomes meaningful only relative to a third term.

Thus:

$$\text{Interaction}(A, B) \neq \text{Gap}(A, B; \cdot).$$

5.2. Meaning requires frame

If N is a stabilized pattern and F is a frame, then:

$$M = F(N).$$

Therefore:

$$\text{same name} \not\Rightarrow \text{same meaning}.$$

5.3. Memory as resonant persistence

Memory is not a passive archive. A pattern is in memory to the degree that it persists under repeated resonance:

$$m_{t+1} = m_t + \alpha \cdot \text{resonate}(m_t, x_t).$$

5.4. Forgetting as decay

Forgetfulness is the weakening of persistence under decay:

$$m_{t+1} = m_t - \beta \cdot \text{erode}(m_t, y_t).$$

5.5. Identity as relational coherence

Identity is not a tag. It is the stability of selected relations across interaction, time, and frame. Identity requires both:

- Dung for persistence through resonance,
- Gap for contrast and self/non-self distinction.

Chapter 6

Existence, Entanglement, and Space

6.1. Existence is not a prime

Proposition 6.1. Existence is not a separate prime. It is a sustained oscillatory condition between resonance and decay.

More precisely:

- Dung provides coherence,
- Hoai provides erosion,
- Sinh reopens novelty,
- existence is the viable regime in which these do not annihilate each other.

6.2. Existence as energy–matter oscillation

Define:

$$\rho = \frac{\text{Dung}}{\text{Hoai} + \text{Sinh}}.$$

This ratio should be interpreted as:

$$\text{resonant organization}/(\text{material decay} + \text{novelty pressure}).$$

The living window is:

$$\theta_{\min} < \rho < \theta_{\max}.$$

This can be stated in plain language:

> Existence is the condition in which energetic resonance organizes matter strongly enough to sustain structure, while matter still remains vulnerable enough to allow transformation.

6.3. The subtle point about stable structure

This is the place most likely to become ambiguous, so it must be stated carefully.

Dung and Hoai are base properties. But in concrete operation:

- energy tends to express Dung more strongly,
- matter tends to express Hoai more clearly.

When energy interacts through time, it generates interference patterns, resonant channels, or stable basins. Matter follows these channels and becomes organized into structure. Thus, the stability of matter is not simply the essence of matter itself. It is a downstream effect of Dung acting through energetic interaction across time.

At the same time, matter once organized becomes the site where Hoai appears most clearly. Why? Because material particles or substructures can collide, push, displace, and stress the positions of the currently stable configuration. Small perturbations may only cause oscillation. Larger perturbations may shift the structure. Strong enough perturbations may break it entirely.

So the correct ontology is:

- Dung organizes,
- Hoai disturbs,
- stable structure is where Dung is temporarily winning over Hoai.

6.4. Existence health score

Let:

$$d = \text{Dung count}, \qquad h = \text{Hoai count}, \qquad s = \text{Sinh count}.$$

Then:

$$\rho = \frac{d}{h + s + \varepsilon}.$$

A practical health score is:

$$\text{health} = \begin{cases} \frac{\rho}{\theta_{\min}}, & \rho < \theta_{\min}, \\ \frac{\theta_{\max}}{\rho}, & \rho > \theta_{\max}, \\ 1 - \dfrac{|\log(\rho)|}{\log(\theta_{\max})}, & \theta_{\min} \le \rho \le \theta_{\max}. \end{cases}$$

6.5. Entanglement is not a prime

Proposition 6.2. Entanglement is not a prime interaction. It is a conservation consequence of pair creation under Sinh.

If Sinh creates a pair (A, B) with:

$$\text{spin}(A) + \text{spin}(B) = 0,$$

then the complementary relation is already present. Later collapse only reveals it.

6.6. Space on the material side

Space is better understood through exclusion. Same-like states that cannot fully occupy the same effective condition must remain apart:

$$\text{Space} = \text{material separation induced by exclusion.}$$

Dung does not create that separation. Rather:

- exclusion creates room,
- resonance moves through the room,
- stable structure forms where resonance successfully channels matter.

Chapter 7

Minimal Formal Sketch and System Mapping

7.1. Interaction selector

Let $A, B \in \mathcal{Q}$. Then:

$$\mathcal{I}(A,B) = \begin{cases} \mathrm{Sinh}(A,B), & \mathrm{collapsed}(A) = 0,\ \mathrm{collapsed}(B) = 0, \\ \mathrm{Dan}(A,B), & \mathrm{collapsed}(A) \neq \mathrm{collapsed}(B), \\ \mathrm{Dung}(A,B), & \mathrm{collapsed}(A) = \mathrm{collapsed}(B) = 1 \text{ and } \mathrm{compat}(A,B) \geq \tau, \\ \mathrm{Hoai}(A,B), & \mathrm{collapsed}(A) = \mathrm{collapsed}(B) = 1 \text{ and } \mathrm{compat}(A,B) < \tau. \end{cases}$$

7.2. Compatibility

A simple abstract compatibility function:

$$\mathrm{compat}(A,B) = \lambda_1 \cos(\mathrm{phase}(A) - \mathrm{phase}(B)) + \lambda_2 s(A,B) + \lambda_3 \cos(\mathrm{vec}(A), \mathrm{vec}(B)).$$

7.3. Implementation mapping

A practical personal AI built from this substrate needs:

1. a **State Engine**,
2. an **Interaction Engine**,
3. a **Frame Router**,
4. a **Memory Graph**,
5. a **Decay Engine**,
6. an **Identity Layer**,
7. a **Semantic Projector**,
8. a **Mode Controller**.

7.4. Minimal pseudocode

Conceptual pseudocode

```
class State:
    def __init__(self, spin, phase, vec, collapsed, metadata=None):
        self.spin = spin
        self.phase = phase
        self.vec = vec
        self.collapsed = collapsed
        self.metadata = metadata or {}

def compatibility(A, B, w1=0.4, w2=0.2, w3=0.4):
    phase_term = cos(A.phase - B.phase)
    spin_term  = spin_compat(A.spin, B.spin)
    vec_term   = cosine(A.vec, B.vec)
    return w1 * phase_term + w2 * spin_term + w3 * vec_term

def interact(A, B, tau=0.5):
    if not A.collapsed and not B.collapsed:
        return "SINH"
    if A.collapsed != B.collapsed:
        return "DAN"
    if compatibility(A, B) >= tau:
        return "DUNG"
    return "HOAI"

def gap(A, B, frame):
    a = frame.project(A)
    b = frame.project(B)
    return norm(a - b)

def nmf(entity, frame):
    name = stable_pattern(entity)
    meaning = frame.project(name)
    return {"name": name, "meaning": meaning, "frame": frame}

def existence_health(dung_count, hoai_count, sinh_count,
                     theta_min=0.2, theta_max=5.0):
    ratio = dung_count / (hoai_count + sinh_count + 1e-9)
    if ratio < theta_min:
        return {"state": "DISSOLVING", "recommendation": "STABILIZE"}
    if ratio > theta_max:
        return {"state": "CRYSTALLIZING", "recommendation": "GROW"}
    return {"state": "ALIVE", "recommendation": "TRANSITION"}
```

Part II

Prime Chakra: A Primordial Semantic Anchor Architecture

Chapter 8

Introduction to the Primordial Scaffold

If Part I answers the question, *what kind of world does* PETE *think in?*, then Part II answers a more primitive question: *before a large semantic space exists, what does semantic growth hold onto?*

A personal AI should not begin semantic development on a perfectly flat lexical plane. It needs rootedness. PRIME CHAKRA provides that rootedness through a small set of primordial maps. These are not claimed as religious truth. They are semantic anchor domains.

The function of this part is not to define the whole semantic engine. It is to define the earliest scaffold:

- a small set of anchor maps,
- a set of initial seeds,
- a pairwise relation system,
- a friction-based learning logic,
- a fractal depth structure.

Chapter 9

Why a Primordial Anchor Layer Is Needed

9.1. The flat-space problem

If a personal AI begins from a flat symbol surface, then early learning lacks primitive orientation. Words may co-occur, but co-occurrence alone does not say what kind of domain a word belongs to. This creates:

- ungrounded early meaning,
- weak developmental continuity,
- blurred experiential modes,
- semantic drift without root pressure.

9.2. The design response

The response is to create a small rooted scaffold:

- each new word receives a resonance profile across fixed maps,
- every word pair has a directed, multi-map relation,
- repeated co-occurrence strengthens routes,
- unused routes decay,
- habits emerge from repeated traversals.

Chapter 10

The Eight Primordial Maps

10.1. Map inventory

The primordial scaffold contains eight maps:

1. Root
2. Sacral
3. Solar
4. Heart
5. Throat
6. Third Eye
7. Crown
8. Unconscious

10.2. Interpretive orientation

Map	Primary semantic orientation
Root	survival, grounding, body, stability, base condition
Sacral	emotion, flow, affect, pleasure, creativity
Solar	will, force, action, confidence, directed energy
Heart	connection, attachment, openness, care, bonding
Throat	language, expression, truth, articulation, speech
Third Eye	insight, intuition, pattern, interpretation, vision
Crown	silence, totality, awareness, stillness, unity
Unconscious	aggregate background field built from all seed inventories

Chapter 11

Three-Layer Fractal Structure

11.1. Layer definition

The primordial scaffold is organized into:

$$L1 = \text{Plane}, \qquad L2 = \text{Cluster}, \qquad L3 = \text{Realm}.$$

11.2. Chakra maps

Each of the seven visible chakra maps contains 9 seeds arranged as:

$$9 \text{ seeds } \Rightarrow 3 \text{ planes } \Rightarrow 1 \text{ cluster}.$$

11.3. Unconscious map

The unconscious map aggregates the whole seed field and extends more deeply:

- 21 planes,
- 7 clusters,
- 2 realms.

11.4. Meaning of the layers

- Plane: near-surface recognition,
- Cluster: sub-threshold but influential grouping,
- Realm: deeper background organization.

Chapter 12

Seed Vocabulary

Map	Seed set
Root	đt, sng, s, an, thân, n, gc, sinh, nn
Sacral	cm, xúc, sáng, to, nc, vui, đam, mê, chy
Solar	ý, chí, sc, mnh, t, tin, la, hành, đng
Heart	yêu, đng, tha, gió, kt, ni, tình, thng, m
Throat	nói, nghe, tht, biu, đt, ngôn, ng, li, ting
Third Eye	trc, giác, hiu, nhìn, nghĩa, tng, trí, sáng, sut
Crown	thc, toàn, im, lng, vũ, tr, ng, vô, tch
Unconscious	aggregate of all visible seeds

Chapter 13

Anchor Scoring and Pairwise Learning

13.1. Anchor score

For a word x, define an 8-dimensional anchor profile:

$$A(x) = \big(a_{\text{Root}}(x), a_{\text{Sacral}}(x), a_{\text{Solar}}(x), a_{\text{Heart}}(x), a_{\text{Throat}}(x), a_{\text{ThirdEye}}(x), a_{\text{Crown}}(x), a_{\text{Unconscious}}(x)\big).$$

This profile locates the word relative to the primordial maps.

13.2. Directed pair structure

If the vocabulary contains N words, then the total number of possible directed edges is:

$$N(N-1).$$

Each directed edge (A, B) carries an 8-dimensional weight vector:

$$w_{A\to B} = \big(w_{\text{Root}}, w_{\text{Sacral}}, w_{\text{Solar}}, w_{\text{Heart}}, w_{\text{Throat}}, w_{\text{ThirdEye}}, w_{\text{Crown}}, w_{\text{Unconscious}}\big).$$

13.3. Initialization

For a new word X and an existing word Y:

$$w^{(0)}_{X\to Y,k} = a_k(X)a_k(Y).$$

Chapter 14

Friction-Based Learning

14.1. Core principle

Axiom 14.1 (Friction learning axiom)**.** Repeated co-occurrence activates paths; activated paths strengthen; inactive paths decay.

14.2. Local reinforcement

If A and B appear together:

$$w^{(t+1)}_{A\to B,k} = w^{(t)}_{A\to B,k} + \eta \cdot r_k(A, B),$$

where η is the friction bump and r_k is resonance contribution in map k.

14.3. Global decay

For all edges:

$$w^{(t+1)}_{A\to B,k} = (1-\delta)\, w^{(t)}_{A\to B,k}.$$

Typical defaults:

$$\texttt{FRICTION} = 0.1, \qquad \texttt{DECAY} = 0.002, \qquad \texttt{MIN_W} = 0.001, \qquad \texttt{MAX_W} = 5.0.$$

14.4. Habit formation

Over time:

- repeated paths become habits,
- rarely used paths fade,
- background pressure accumulates in the unconscious map.

Chapter 15

Developmental Rationale and Corpora

15.1. Why a child-like start matters

The primordial layer is meant to resemble early semantic growth, not mature semantic fluency. A child does not begin with massive abstraction. It begins with repeated contact, affective gradients, bodily reference, action patterns, and socially reinforced utterances.

15.2. Corpus directions

Suitable primordial corpora include:

- Vietnamese subtitle dialogue,
- simple Vietnamese sentence corpora,
- CHILDES child-language corpora.

15.3. Why this matters for Pete

Pete is not intended to be only a language predictor. It is meant to become a personal AI with a growth history. Prime Chakra gives it a small, interpretable beginning.

Chapter 16

Technical Architecture for the Primordial Layer

16.1. Folder logic

```
Pete_imagination/
 primordial_pete4/
    chakra_seeds.py
    word_anchor.py
    pairwise_map.py
    fractal_layers.py
    primordial.py
    primordial_map.json
 I_layer/
 I_primordial.py
 ui/
     app.py
     index.html
```

16.2. Minimal pseudocode

```
FRICTION = 0.1
DECAY    = 0.002
MIN_W    = 0.001
MAX_W    = 5.0

MAPS = [
    "root", "sacral", "solar", "heart",
    "throat", "third_eye", "crown", "unconscious"
]

def anchor_scores(word, seeds_by_map):
    scores = []
    for map_name in MAPS:
        score = resonance(word, seeds_by_map[map_name])
        scores.append(score)
```

```
    total = sum(scores) + 1e-9
    return [s / total for s in scores]

def init_edge(ax, ay):
    return [ax[i] * ay[i] for i in range(8)]

def add_word(new_word, vocab, seeds_by_map, graph):
    a_new = anchor_scores(new_word, seeds_by_map)
    for old_word in vocab:
        a_old = anchor_scores(old_word, seeds_by_map)
        graph[(new_word, old_word)] = init_edge(a_new, a_old)
        graph[(old_word, new_word)] = init_edge(a_old, a_new)
    vocab.add(new_word)
```

Part III

Semantic Space Engine: From Raw Text to Living Semantic Space

Chapter 17

Introduction to the Semantic Runtime

If Part I defines the ontology and Part II defines the primordial anchors, Part III defines the machinery that turns raw text into a living semantic space.

The runtime is built around a strong design decision:

> PETE should build meaning from experience rather than inherit a fixed semantic geometry.

The semantic engine therefore does not begin by looking up a static embedding. It begins by reading the situation of an input, measuring symbol behavior, building co-occurrence structure, verifying nodes, updating meanings through field gravity, forming pre-linguistic primordials, and only then producing a dynamic context-conditioned representation.

Chapter 18

Pipeline Overview

The full pipeline is:

1. normalize text,
2. detect boundary and language,
3. segment symbols,
4. extract situation signal before tokenization,
5. measure symbol statistics,
6. build a LearningEvent,
7. update co-occurrence,
8. compute PPMI,
9. verify nodes,
10. tick the P1 lifecycle,
11. update meaning by field gravity,
12. build primordials,
13. compute DCEE.

Pipeline summary

Raw text

$\longrightarrow$ normalize $\longrightarrow$ boundary + language boost $\longrightarrow$ segment $\longrightarrow$ situation $\longrightarrow$ measure $\longrightarrow$ LearningEve

Chapter 19

Preprocessing and Situation-First Logic

19.1. Normalization

Raw text is first normalized:

$$x \mapsto \text{normalize}(x).$$

19.2. Boundary and language boost

Language is detected early and converted into a small field prior:

- Vietnamese may increase language and intuition pressure,
- English may increase language and logic pressure,
- Chinese may increase visual and intuition pressure.

19.3. Segmentation

Segmentation is language-aware:

- whitespace or syllable-like segmentation for Vietnamese,
- whitespace segmentation for English,
- character segmentation for Chinese.

19.4. Situation before tokenization

The engine reads how something is said before reading what was said. Situation signal includes:

- intensity,
- urgency,
- emotional charge,
- interrogative or assertive force,
- social pressure.

This is converted into:

$$\mathbf{R}_{situation} \in \mathbb{R}^6,$$

over the canonical ODFS fields:

$$\{\text{emotion}, \text{logic}, \text{reflection}, \text{visual}, \text{language}, \text{intuition}\}.$$

A typical initial blend is:

$$\mathbf{R}_0 = w_s \mathbf{R}_{situation} + w_n \mathbf{R}_{node} + w_t \mathbf{R}_{stream} + w_p \mathbf{R}_{prim},$$

with defaults:

$$w_s = 0.40, \qquad w_n = 0.40, \qquad w_t = 0.15, \qquad w_p = 0.05.$$

Chapter 20

Symbol Statistics and LearningEvent

20.1. Symbol statistics

Each symbol is measured through:

- frequency $f(n)$,
- contextual frequency $\mathrm{CF}(n)$,
- entropy $D(n)$,
- surprise $\mathrm{Surp}(n)$.

These distinguish:

- frequent but semantically flat symbols,
- contextually rich symbols,
- rare but surprising symbols,
- unstable polysemous symbols.

20.2. LearningEvent contract

A LearningEvent contains:

- normalized symbols,
- context pairs,
- source metadata,
- language hint and boost,
- situation vector,
- CF, entropy, and surprise,
- source weight.

Source weights obey:

$$w_{\text{user}} = 1.0, \qquad w_{\text{corpus}} = 0.8, \qquad w_{\text{pete}} = 0.5.$$

Chapter 21

Co-occurrence Structure and Node Verification

21.1. Context window

Let the context radius be:

$$r = 3.$$

For each pair (a, b) in the local window:

$$C(a, b) \leftarrow C(a, b) + 1.$$

21.2. PMI and PPMI

$$\mathrm{PMI}(a, b) = \log_2 \frac{P(a, b)}{P(a)P(b)}, \qquad \mathrm{PPMI}(a, b) = \max(0, \mathrm{PMI}(a, b)).$$

Where:

$$P(a, b) = \frac{C(a, b)}{T \cdot r \cdot 2}, \qquad P(a) = \frac{C(a)}{T}, \qquad P(b) = \frac{C(b)}{T}.$$

21.3. Node strength

The strength function is:

$$L(n) = f(n)^{0.5} \cdot \mathrm{CF}(n)^{1.0} \cdot D(n)^{0.8} \cdot (1 + \mathrm{Surp}(n))^{0.3} \cdot w_{\mathrm{source}}.$$

A typical threshold:

$$\theta_L = 1.5.$$

21.4. Coherence

A separate stability measure is:

$$\mathrm{coh}(n) = 1 - \frac{\mathrm{std}(\Delta m_n)}{\mathrm{mean}(\Delta m_n) + \varepsilon}.$$

A typical threshold:

$$\theta_{\mathrm{coh}} = 0.1.$$

A node is promoted only if both conditions hold:

$$L(n) \geq \theta_L, \qquad \mathrm{coh}(n) \geq \theta_{\mathrm{coh}}.$$

Chapter 22

Field Gravity and P1 Lifecycle

22.1. Canonical ODFS fields

The engine uses six canonical fields:

$$\{\text{emotion}, \text{logic}, \text{reflection}, \text{visual}, \text{language}, \text{intuition}\}.$$

Initial node meaning is uniform:

$$m_0(n) = \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right).$$

22.2. Field gravity

If F_j is a field and $N(n)$ is the node neighborhood:

$$g_j(n) = \frac{|N(n) \cap M(F_j)|}{|N(n)| + \varepsilon}.$$

Meaning is updated by:

$$m_j(n) \leftarrow m_j(n) + \eta_f \, w_s \, (g_j(n) + b_j),$$

then normalized.

22.3. Light P2 pull

For mature nodes:

$$m(n) \leftarrow (1 - \lambda) m(n) + \lambda m_{P2}, \qquad \lambda = 0.005.$$

22.4. P1 lifecycle

Nodes evolve through:

$$\text{Vo} \to \text{Sinh} \to \text{Dan} \to\to \text{Dung} \to \text{Hoai}.$$

Thresholds:

$$T_{\text{fire}} = 3, \qquad T_{\text{field}} = 6, \qquad T_{\text{core}} = 20.$$

The important clarification from Part I still holds here:

- Dung-phase at runtime means stable persistence *emerging from repeated resonance*,
- Hoai-phase at runtime means structural weakening *under decay and displacement*.

Chapter 23

Pre-linguistic Primordials and DCEE

23.1. Why primordials are needed

Not all meaning is fully nameable. Some structures form around a hub but still contain unresolved residue.

23.2. Hub and satellites

A primordial forms when a hub has at least:

$$3 \text{ satellites.}$$

If the hub is h and satellite vectors are $v_1, \ldots, v_k$:

$$c_h = \frac{1}{k} \sum_{i=1}^{k} v_i.$$

Gap:

$$g_h = 1 - \cos(m(h), c_h).$$

Unnamed feel:

$$u_h = c_h - m(h).$$

23.3. Dynamic context embedding

The runtime embedding is not a lookup. It is recomputed:

$$\mathbf{R}_{ctx} = 0.6\, m(n) + 0.4\, c_{\text{local}},$$

then:

$$\mathbf{R} = \mathbf{R}_{ctx} + 0.2\, o + 0.1\, p2.$$

A fuller blend can include:

$$\mathbf{R} = \lambda_0 \mathbf{R}_0 + \lambda_1 m(n) + \lambda_2 c_{\text{local}} + \lambda_3 o + \lambda_4 p2.$$

This is the DCEE principle:

> the same symbol may map to different vectors under different contexts.

Chapter 24

Semantic Exclusion and Anti-Collapse

Part I defined space as structured separation under exclusion. The semantic runtime therefore also needs a distinction-preserving mechanism. Otherwise, pure resonance would collapse nearby concepts into one basin.

A semantic analogue of exclusion is:

> if two semantic nodes become too similar, the system must preserve their distinction rather than allowing total collapse.

24.1. Semantic Pauli

If the meaning similarity between node A and node B is too high, the system slightly nudges them apart along the dimension of maximal difference.

```
def semantic_pauli(node_A, node_B, ODFS_FIELDS, exclusion_threshold=0.95):
    vec_A = [node_A.meaning[f] for f in ODFS_FIELDS]
    vec_B = [node_B.meaning[f] for f in ODFS_FIELDS]
    similarity = cosine(vec_A, vec_B)

    if similarity > exclusion_threshold:
        diff = [node_A.meaning[f] - node_B.meaning[f] for f in ODFS_FIELDS]
        max_diff_idx = max(range(len(diff)), key=lambda i: abs(diff[i]))
        max_diff_field = ODFS_FIELDS[max_diff_idx]

        node_A.meaning[max_diff_field] *= 1.05
        node_B.meaning[max_diff_field] *= 0.95

        for node in [node_A, node_B]:
            total = sum(node.meaning.values()) + 1e-9
            for k in node.meaning:
                node.meaning[k] /= total
```

This protects conceptual plurality against total resonant collapse.

Chapter 25

Runtime Data Structures and Parameters

25.1. Core stores

The semantic runtime writes to:

1. `node_store`,
2. `graph`,
3. `plane_store`,
4. `identity_store`.

25.2. Defaults

Useful defaults include:

$$\text{window radius} = 3, \quad \theta_L = 1.5, \quad \theta_{\text{coh}} = 0.1.$$

$$T_{\text{fire}} = 3, \quad T_{\text{field}} = 6, \quad T_{\text{core}} = 20.$$

$$\text{min satellites} = 3, \quad \theta_{gap} = 0.02.$$

Chapter 26

Unified Interpretation Across the Three Parts

The trilogy now forms one coherent stack:

Part I — Ontology

Defines:

- state,
- prime interaction,
- Dung/Hoai asymmetry,
- difference,
- frame,
- existence,
- space.

Part II — Primordial Anchoring

Defines:

- earliest semantic maps,
- seed words,
- pairwise friction learning,
- rooted developmental scaffold.

Part III — Semantic Runtime

Defines:

- raw text processing,

- node emergence,
- co-occurrence structure,
- field gravity,
- primordials,
- dynamic embedding.

Chapter 27

Conclusion

This monograph presented a unified internal foundation for building a personal AI system called Pete. The three parts were designed to work together.

Part I defined an interaction-based ontology in which states and primes replace static object-first thinking. It also clarified the most delicate ontological point in the framework: Dung and Hoai are base properties, but they do not manifest identically across reality. Dung is the resonant side, and energy tends to manifest it more strongly. Hoai is the decaying side, and matter tends to manifest it more clearly. Stable material structure is not the essence of matter, but the temporary result of matter being guided into resonant grooves formed by energetic interaction through time. Yet the same material embodiment becomes vulnerable to displacement, collision, and breakdown, which is why Hoai appears clearly in matter.

Part II provided a primordial semantic scaffold. It ensured that early semantic growth does not begin on an unrooted plane, but on a small set of interpretable anchor maps whose pairwise habits can accumulate over time.

Part III operationalized the whole system. It showed how raw text can become a living semantic space through preprocessing, situation-first logic, co-occurrence structure, node verification, field gravity, primordials, and dynamic context embedding.

The framework remains internal, provisional, and incomplete. But it offers something practically useful: a way to design a personal AI whose semantics, memory, identity, and continuity arise from structured interaction rather than from frozen inherited geometry.

Appendix A: Prime Summary Table

No.	Prime	Meaning
0	Vo	latent potential under non-interaction
1	Sinh	emergence, novelty, unresolved generation
2	Dan	probing, testing, pre-collapse evaluation
3	Dung	resonance, coherence, constructive coupling
4	Hoai	decay, erosion, displacement, breakdown
5	Gap	measured difference under frame
6	Time	ordered interaction sequence
7	Space	structured separation under exclusion
8	NMF	named meaning in frame

Appendix B: Core Equations

$$\mathrm{Gap}(A, B; F) = F(A) - F(B).$$

$$\mathrm{NMF}(x, F) = \big(N(x),\ F(N(x)),\ F\big).$$

$$\rho = \frac{\mathrm{Dung}}{\mathrm{Hoai} + \mathrm{Sinh}}.$$

$$\theta_{\min} < \rho < \theta_{\max}.$$

$$L(n) = f(n)^{0.5} \cdot \mathrm{CF}(n)^{1.0} \cdot D(n)^{0.8} \cdot (1 + \mathrm{Surp}(n))^{0.3} \cdot w_{\mathrm{source}}.$$

$$\mathrm{coh}(n) = 1 - \frac{\mathrm{std}(\Delta m_n)}{\mathrm{mean}(\Delta m_n) + \varepsilon}.$$

$$\mathrm{PMI}(a, b) = \log_2 \frac{P(a, b)}{P(a)P(b)}, \qquad \mathrm{PPMI}(a, b) = \max(0, \mathrm{PMI}(a, b)).$$

$$\mathbf{R}_{ctx} = 0.6\, m(n) + 0.4\, c_{\mathrm{local}}, \qquad \mathbf{R} = \mathbf{R}_{ctx} + 0.2\, o + 0.1\, p2.$$

Appendix C: Minimal Unified Pseudocode

```
class State:
    def __init__(self, spin, phase, vec, collapsed, metadata=None):
        self.spin = spin
        self.phase = phase
        self.vec = vec
        self.collapsed = collapsed
        self.metadata = metadata or {}

def interact(A, B, tau=0.5):
    if not A.collapsed and not B.collapsed:
        return "SINH"
    if A.collapsed != B.collapsed:
        return "DAN"
    if compatibility(A, B) >= tau:
        return "DUNG"
    return "HOAI"

def existence_health(dung_count, hoai_count, sinh_count,
                     theta_min=0.2, theta_max=5.0):
    ratio = dung_count / (hoai_count + sinh_count + 1e-9)
    if ratio < theta_min:
        return {"state": "DISSOLVING", "recommendation": "STABILIZE"}
    if ratio > theta_max:
        return {"state": "CRYSTALLIZING", "recommendation": "GROW"}
    return {"state": "ALIVE", "recommendation": "TRANSITION"}

def anchor_scores(word, seeds_by_map):
    scores = []
    for map_name in seeds_by_map:
        scores.append(resonance(word, seeds_by_map[map_name]))
    total = sum(scores) + 1e-9
    return [s / total for s in scores]

def L_of_n(freq, CF, D, Surp, source_weight):
    return (freq ** 0.5) * (CF ** 1.0) * (D ** 0.8) * ((1 + Surp) ** 0.3) *
     source_weight

def compute_ppmi(a, b, cooc, total_tokens, vocab, window=3):
```

```
    count_ab = cooc.get((a, b), 0)
    if count_ab == 0:
        return 0.0
    count_a = sum(cooc.get((a, x), 0) for x in vocab)
    count_b = sum(cooc.get((x, b), 0) for x in vocab)
    denom = total_tokens * window * 2
    P_ab = count_ab / max(denom, 1)
    P_a  = count_a / max(total_tokens, 1)
    P_b  = count_b / max(total_tokens, 1)
    pmi  = math.log2(P_ab / (P_a * P_b + 1e-9) + 1e-9)
    return max(0.0, pmi)

def dcee(node, local_context_vec, odfs_mod, p2_pull):
    base = 0.6 * node.vec + 0.4 * local_context_vec
    return base + 0.2 * odfs_mod + 0.1 * p2_pull
```